Lab Report 08

Course title: Computer Network Laboratory
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Submitted to-

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Title: Implementation of VLSM Subnetting

Objective:

- To understand the concept of Variable Length Subnet Masking (VLSM).
- To divide a network into subnets of different sizes according to host requirements.
- To use IP addresses efficiently and reduce address wastage.

Description:

The network 192.168.20.0/24 was divided into four subnets using VLSM according to the host requirements of ICT, CSE, ES and DS.

Department	Required hosts	Network address	Prefix	Subnet mask	Usable host range	Broadcast
ICT	100	192.168.20.0	/25	255.255.255.128	.1 – .126	.127
CSE	45	192.168.20.128	/26	255.255.255.192	.129 – .190	.191
ES	20	192.168.20.192	/27	255.255.255.224	.193 – .222	.223
DS	10	192.168.20.224	/28	255.255.255.240	.225 – .238	.239

Two Cisco 2911 routers, four switches and eight PCs were added. Each router was connected to two switches and each switch was connected to two PCs. R1 was used for the ICT and CSE departments, while R2 was used for the ES and DS departments.

IP addresses and subnet masks were configured on the router interfaces. The two routers were connected using the 192.168.20.240/30 subnet. Static routes were configured on both routers to allow communication between all four departmental networks. Each PC was assigned an IP address, subnet mask and default gateway according to its subnet.

Connection:

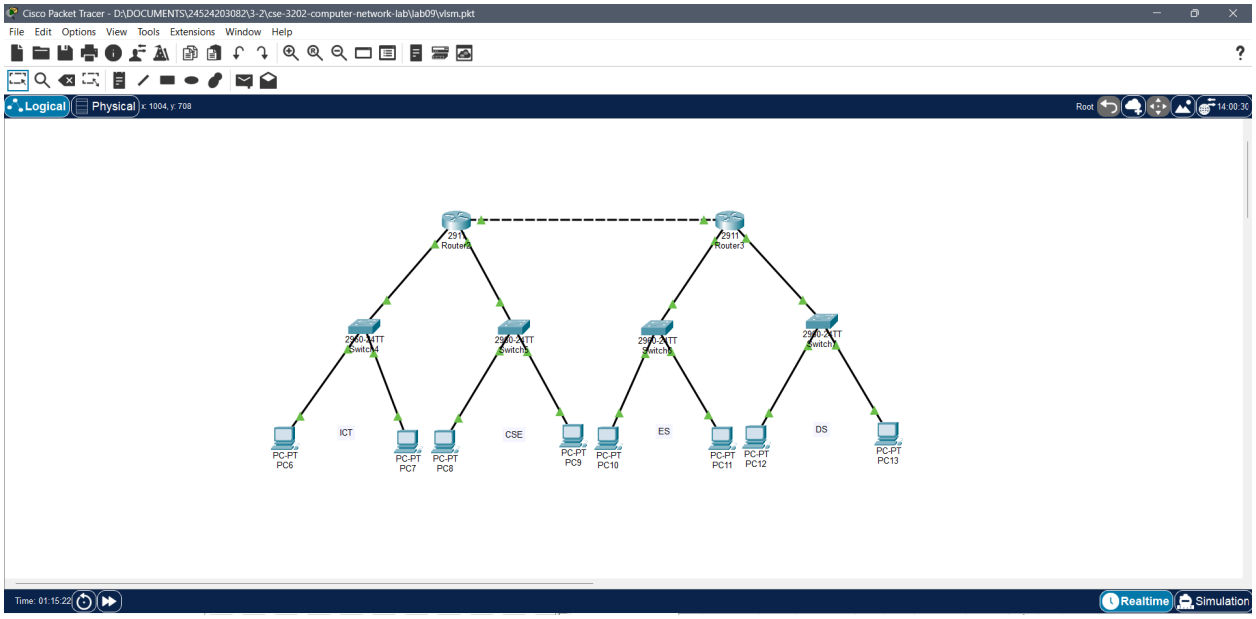


Figure 1: Network Topology

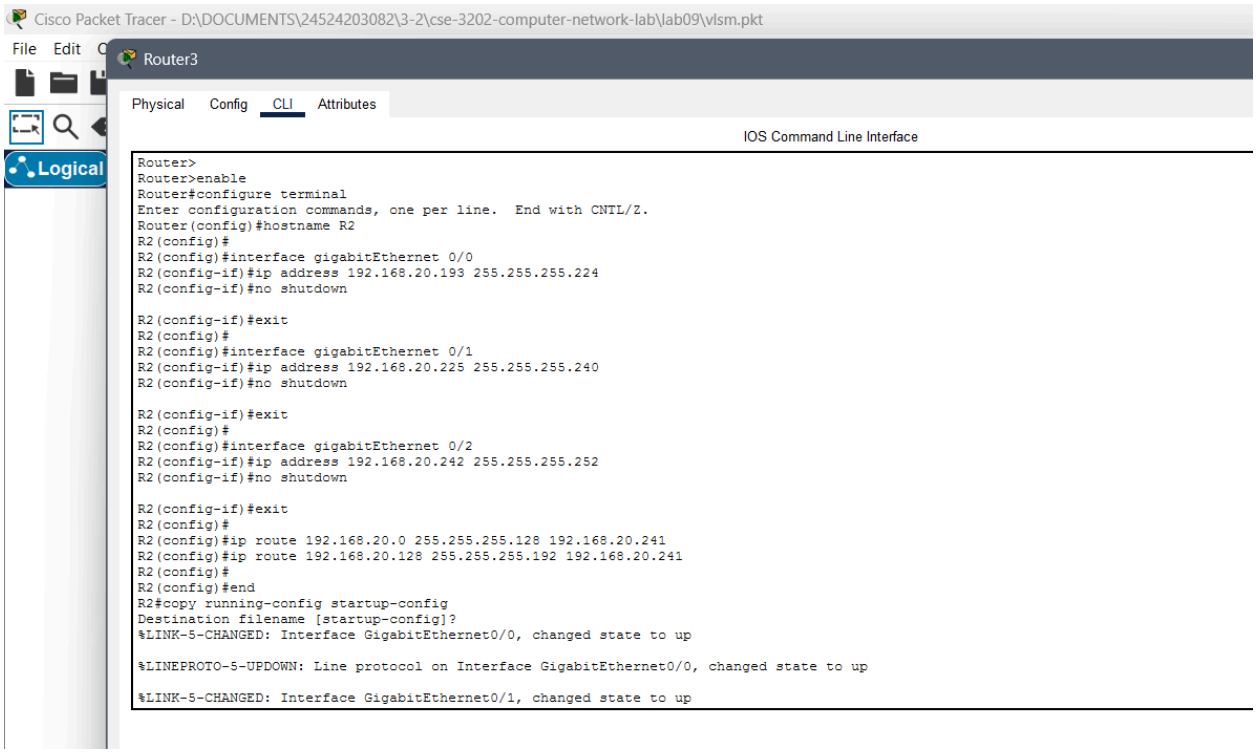


Figure 2: Router CLI

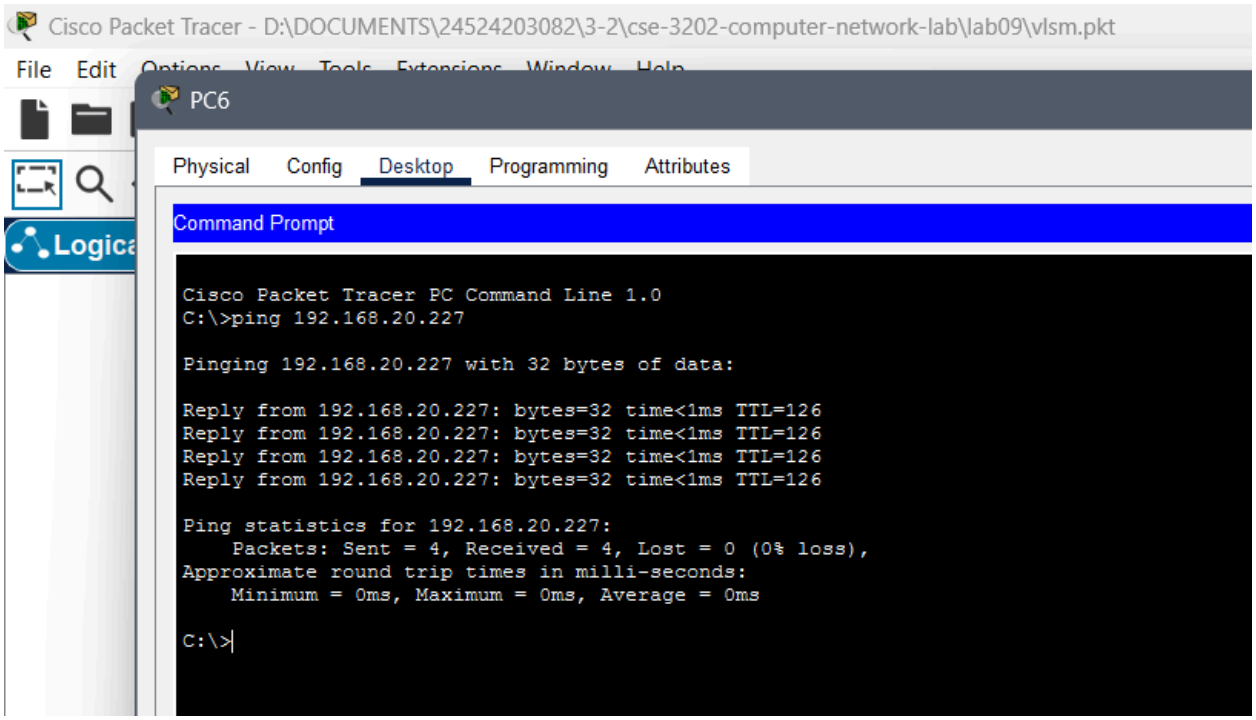


Figure 3: Ping Connectivity Test Between Different LANs